

[illegible]

1. Work the algorithm

WHAT YOU SEE

Banner: WORK THE ALGORITHM — DO NOT JUMP TO SIADH

WHAT IT ENCODES

Hyponatremia ($\text{Na} < 135$) demands a fixed stepwise workup, because SIADH is a diagnosis of EXCLUSION. The sequence is: serum osmolality → exclude mimics (pseudohyponatremia, hyperglycemia) → urine osmolality → urine sodium → clinical volume status. Each step narrows the differential; jumping straight to SIADH skips treatable mimics like adrenal insufficiency and hypothyroidism.

BOARD TRAP

WRONG: labeling a euvolemic hyponatremic patient "SIADH" on first glance. SIADH cannot be diagnosed until hypothyroidism, adrenal/glucocorticoid insufficiency, thiazides, and low-solute states are excluded. The discriminator: SIADH requires normal thyroid/adrenal function and no recent thiazide.

2. Order of steps

WHAT YOU SEE

Bottom banner: OSM then MIMICS then UOSM then UNa then VOLUME

WHAT IT ENCODES

Memorize the order: serum OSM → MIMICS → urine OSM → urine Na → VOLUME. Osmolality first sorts hypotonic (true) from isotonic (pseudo) and hypertonic (glucose/mannitol). Urine osm then tells you whether ADH is on (>100) or off (<100), urine Na localizes renal vs extrarenal, and volume status finalizes the category and treatment.

BOARD TRAP

WRONG: starting with volume status (the classic textbook tree) before checking osmolality. If you skip the osmometer you can mistreat isotonic pseudohyponatremia or hypertonic hyperglycemic hyponatremia as if it were true hypotonic hyponatremia.

3. Osmometer step 1

WHAT YOU SEE

Osmometer gauge labeled OSMOMETER, 275 NORMAL TRUE / HIGH SHIFT zones

WHAT IT ENCODES

Step 1 is always measured serum osmolality (normal $\sim 275\text{--}295$ mOsm/kg). LOW osm (<275) = true hypotonic hyponatremia, the only category needing the full volume-status algorithm. NORMAL osm = isotonic pseudohyponatremia (lab artifact). HIGH osm (>295) = hypertonic hyponatremia (hyperglycemia, mannitol) where osmotically active solute pulls water out of cells, diluting Na.

BOARD TRAP

WRONG: treating every low Na as water excess. If serum osm is normal or high, the problem is not free-water retention — it is a lipid/protein artifact or an osmotic solute. Measure osmolality before committing to any treatment.

4. Pseudo (isotonic)

WHAT YOU SEE

Round figure labeled PSEUDO

WHAT IT ENCODES

Isotonic pseudohyponatremia is a laboratory artifact of older flame-photometry/indirect-ISE methods caused by severe hyperlipidemia (high triglycerides/cholesterol) or paraproteinemia (multiple myeloma, Waldenström). The solid phase displaces plasma water; serum osmolality is NORMAL and there is no true sodium or water disturbance.

BOARD TRAP

WRONG: aggressively correcting a "sodium of 128" in a myeloma or hypertriglyceridemia patient with a normal serum osm. There is no real hyponatremia to treat — the fix is to recognize the artifact (or use a direct ISE/whole-blood gas analyzer), not to give hypertonic saline.

5. Hyperglycemia +1.6/2.4

WHAT YOU SEE

Striped figure labeled 1.6 / 2.4 per 100

WHAT IT ENCODES

Hyperglycemia causes hypertonic (translocational) hyponatremia: glucose draws water into the vascular space, diluting Na. Correct measured Na UP by 1.6 mEq/L (or ~ 2.4 above glucose ~ 400) for every 100 mg/dL glucose over 100. Serum osm is HIGH. Treating the hyperglycemia restores Na.

BOARD TRAP

WRONG: giving hypertonic saline for the "Na 126" in a DKA patient with glucose 600. The corrected Na is actually ~ 134 (near normal); the real treatment is insulin and fluids for the glucose. Forgetting the correction leads to dangerous overtreatment.

6. Cortisol mimic

WHAT YOU SEE

Card labeled CORTISOL among MIMICS

WHAT IT ENCODES

Glucocorticoid (cortisol) deficiency — primary adrenal insufficiency or secondary/central hypopituitarism — causes euvolemic hyponatremia. Low cortisol removes feedback inhibition of CRH/ADH and impairs free-water excretion, mimicking SIADH. Check a morning cortisol and/or cosyntropin stimulation test before diagnosing SIADH.

BOARD TRAP

WRONG: managing adrenal-insufficiency hyponatremia with fluid restriction like SIADH. It will not correct and can precipitate adrenal crisis — the answer is glucocorticoid replacement (hydrocortisone). Primary adrenal insufficiency also gives high K + low Na, a clue SIADH never produces.

7. TSH mimic

WHAT YOU SEE

Card labeled TSH among MIMICS

WHAT IT ENCODES

Severe hypothyroidism (especially myxedema) causes euvolemic hyponatremia via reduced cardiac output and GFR with non-osmotic ADH release. It is a mandatory SIADH mimic to exclude — order TSH and free T4. Unlike adrenal insufficiency, potassium is NORMAL in hypothyroid hyponatremia.

BOARD TRAP

WRONG: diagnosing SIADH without thyroid studies. A myxedematous patient's hyponatremia responds to thyroid hormone replacement, not to fluid restriction alone; SIADH requires documented euthyroid status.

8. K high/Na low = adrenal

WHAT YOU SEE

Card: K HIGH, Na LOW, ADRENAL NOT SIADH

WHAT IT ENCODES

The combination of LOW sodium + HIGH potassium ($\pm$ metabolic acidosis, hypotension, hyperpigmentation) points to PRIMARY adrenal insufficiency (Addison) from combined gluco- and mineralocorticoid (aldosterone) loss. SIADH never causes hyperkalemia — it is a pure water problem with normal K.

BOARD TRAP

WRONG: calling hyperkalemic hyponatremia SIADH. The hyperkalemia is the giveaway for aldosterone deficiency (Addison's) — order cortisol/cosyntropin and give hydrocortisone, do not fluid-restrict.

9. Low urate = SIADH

WHAT YOU SEE

Card: LOW URATE DRY

WHAT IT ENCODES

SIADH characteristically lowers serum uric acid (<4 mg/dL) through dilution and ADH/volume-mediated uricosuria. A low urate supports SIADH; a HIGH urate (>5 – 6) argues for volume depletion/hypovolemia, where the kidney avidly reabsorbs urate. This is a high-yield discriminator between euvolemic SIADH and hypovolemic hyponatremia.

BOARD TRAP

WRONG: ignoring uric acid when distinguishing SIADH from hypovolemia. An elevated urate in a hyponatremic patient should steer you AWAY from SIADH and toward volume depletion (and away from fluid restriction toward isotonic saline).

10. Urine osm gauge

WHAT YOU SEE

Dial: URINE OSM 100 to 400 MIXED, 100 AVP OFF, 400 AVP ON

WHAT IT ENCODES

After confirming hypotonicity, urine osmolality reveals ADH activity. Uosm <100 mOsm/kg = ADH appropriately OFF, dilute urine (primary polydipsia, low-solute beer potomania/tea-and-toast). Uosm >100 = ADH ON, inappropriately concentrated urine (SIADH, hypovolemia, edematous states, thiazides).

BOARD TRAP

WRONG: diagnosing SIADH with a maximally dilute urine. If Uosm is <100 , ADH is suppressed and the problem is excess water intake or low solute — SIADH requires INAPPROPRIATELY concentrated urine (>100) despite hypotonic plasma.

11. Dry / primary polydipsia

WHAT YOU SEE

Figure labeled DRY at upper right

WHAT IT ENCODES

Primary (psychogenic) polydipsia: massive water intake (often >10 L/day, classically in psychiatric patients) overwhelms renal free-water excretion. ADH is correctly suppressed, so urine is maximally dilute (Uosm <100 , often <50 – 60). Treatment is water restriction; correction can be rapid (autocorrection risk).

BOARD TRAP

WRONG: fluid restriction is correct, but beware OVER-rapid autocorrection. Once water intake stops, the kidney rapidly dumps free water and Na can rise too fast — risking osmotic demyelination. Monitor closely and consider DDAVP clamp if Na overcorrects.

12. Euvolemic mid Uosm

WHAT YOU SEE

Figure labeled EUVOLEMIC

WHAT IT ENCODES

A clinically euvolemic patient (no edema, no dryness/orthostasis) with inappropriately concentrated urine (Uosm >100), urine Na >30 , low uric acid, and normal thyroid/adrenal function = SIADH. It is the endpoint of the euvolemic arm AFTER all mimics are excluded.

BOARD TRAP

WRONG: equating "euvolemic + concentrated urine" with SIADH without finishing the exclusion. Hypothyroidism, glucocorticoid deficiency, thiazides, and reset osmostat can all look identical — the diagnosis is only secure once these are ruled out.

13. Urine Na step

WHAT YOU SEE

Sign: URINE Na with 20 SAVING and 30 LEAKING gates

WHAT IT ENCODES

Urine sodium localizes the lesion. Una <20 mEq/L = kidney conserving Na = extrarenal/hypovolemic loss (vomiting, diarrhea, third-spacing) or edematous states (HF, cirrhosis). Una >30 – 40 = renal Na leak (SIADH, diuretics, salt-wasting nephropathy, adrenal insufficiency, RTA).

BOARD TRAP

WRONG: interpreting urine Na in a patient on diuretics. Diuretics force renal Na loss and make Una high regardless of the true cause, masking hypovolemia. Always check the medication list and ideally interpret Una off diuretics.

14. Thiazide mimic 1-2wk

WHAT YOU SEE

Sign: WAIT 1 TO 2 WK, THIAZIDE MIMIC

WHAT IT ENCODES

Thiazides (not loops) classically cause a euvolemic, SIADH-indistinguishable hyponatremia: they impair urinary dilution at the distal tubule while sparing concentrating ability, and frequently affect elderly women within weeks of starting. Management: STOP the thiazide and recheck Na in 1–2 weeks before labeling SIADH.

BOARD TRAP

WRONG: diagnosing SIADH in a patient currently on a thiazide. The thiazide must be discontinued and the patient reassessed — many "SIADH" cases resolve completely once the drug is stopped. Loops, by contrast, impair concentrating ability and rarely cause hyponatremia.

15. Beer / low solute

WHAT YOU SEE

Man with beer labeled LOW SOLUTE LOW INTAKE

WHAT IT ENCODES

Low-solute states — beer potomania and tea-and-toast diet — cause hyponatremia from inadequate dietary solute (protein/salt) limiting free-water excretion despite suppressed ADH. Labs show LOW Uosm (<100–200) and LOW Una. Correction is with solute/protein (and saline if depleted); free water dumps rapidly once solute is given.

BOARD TRAP

WRONG: confusing low-solute hyponatremia with SIADH and fluid-restricting hard. Uosm is LOW here (ADH off), so the mechanism is solute-limited excretion. Beware rapid autocorrection/overcorrection once the patient eats — restrict the rate of Na rise to avoid osmotic demyelination.
